

* Ensemble learning

Ensemble learning means using many models together instead of one model.

In machine learning:

one model = one opinion

ensemble = many opinions combined

Final answer = better and more accurate.

① Different algorithms

Using different types of models to learn the same problem.

We use different algorithms because one model may miss something another model catches.

Example: Predict house price using linear regression, decision tree, random forest, SVR

② Different Training data sets

Training the same algorithm multiple times but on different data samples

Model becomes less dependent on one dataset.

Weak and strong learners.

The main idea is that many weak learners combined to form one strong learner.

Ensemble learning improves performance by combining multiple weak learners trained on different algorithms or different datasets to form a strong learner.

* Ensemble technique

① Bagging

random forest

② Boosting

adaboost - outdated nobody use it in organisation

catboost - outdated

xgboost

lgbm - very powerful works very well for big data

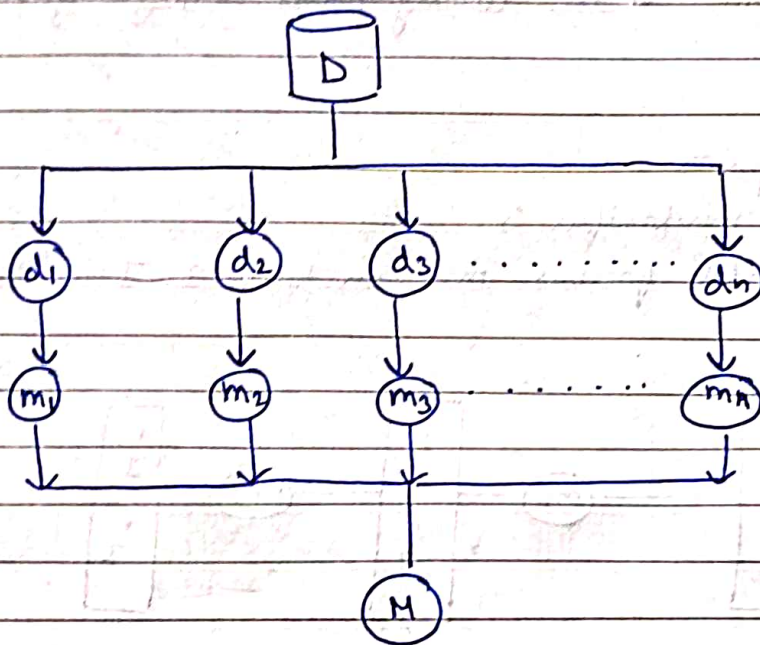
③ Voting or Stacking

Voting will be applied in bagging also voting will apply in boosting also.

* Bagging technique

We bag records from original dataset.

We have original dataset - from original dataset if we bag the records and we create multiple ~~new~~ datasets.



Random forest comes under bagging technique.

What is bagging?

Train many models independently on different random samples of data and then combine their results

- Same algo
- Different training dataset
- Models trained in parallel

Random forest comes under bagging because it trains multiple decision trees independently on different bootstrap samples of the dataset and combines their predictions through voting or averaging.

* Boosting Technique

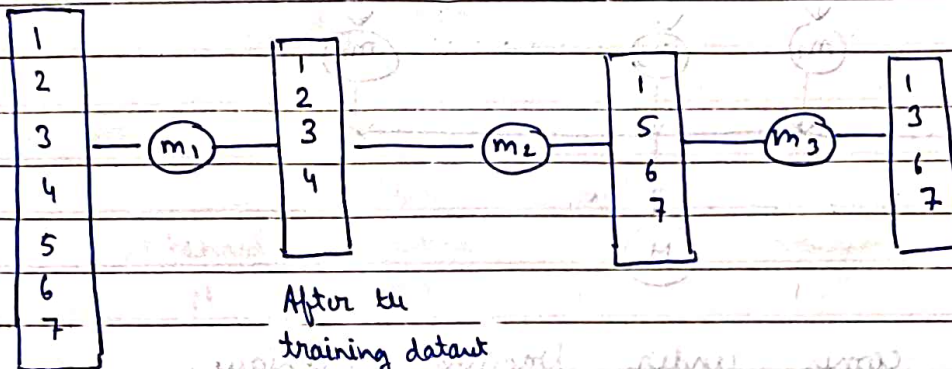
Boosting is a technique where models are trained one after another and each new model focuses on the mistakes of the previous ones.

- ① Train a simple model. It makes mistakes.
- ② Give more importance to the wrong points. They get more attention.

For eg: Total 10 points

- Model 1 misclassifies 3

Next model focuses more on those 3. Tries hard to fix them.



Training
dataset

T_1

After the
training dataset
is passed through

model m_1 . Some

pts are misclassified

Next model will consider

other points as a train

and these pts as a test

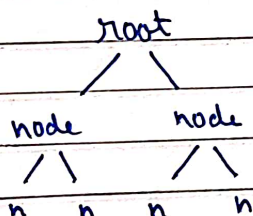
Q What are XGBoost and LGBM?

Both are boosting algos that use decision trees.

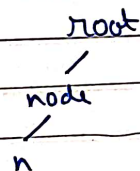
- They build many small trees one after another
- Each new tree fixes the mistakes of earlier trees.

The only difference is how they grow a tree:

XGBoost grows tree level by level



LGBM grows tree leaf by leaf



- LGBM uses less memory and works ^{well} in case of large data.